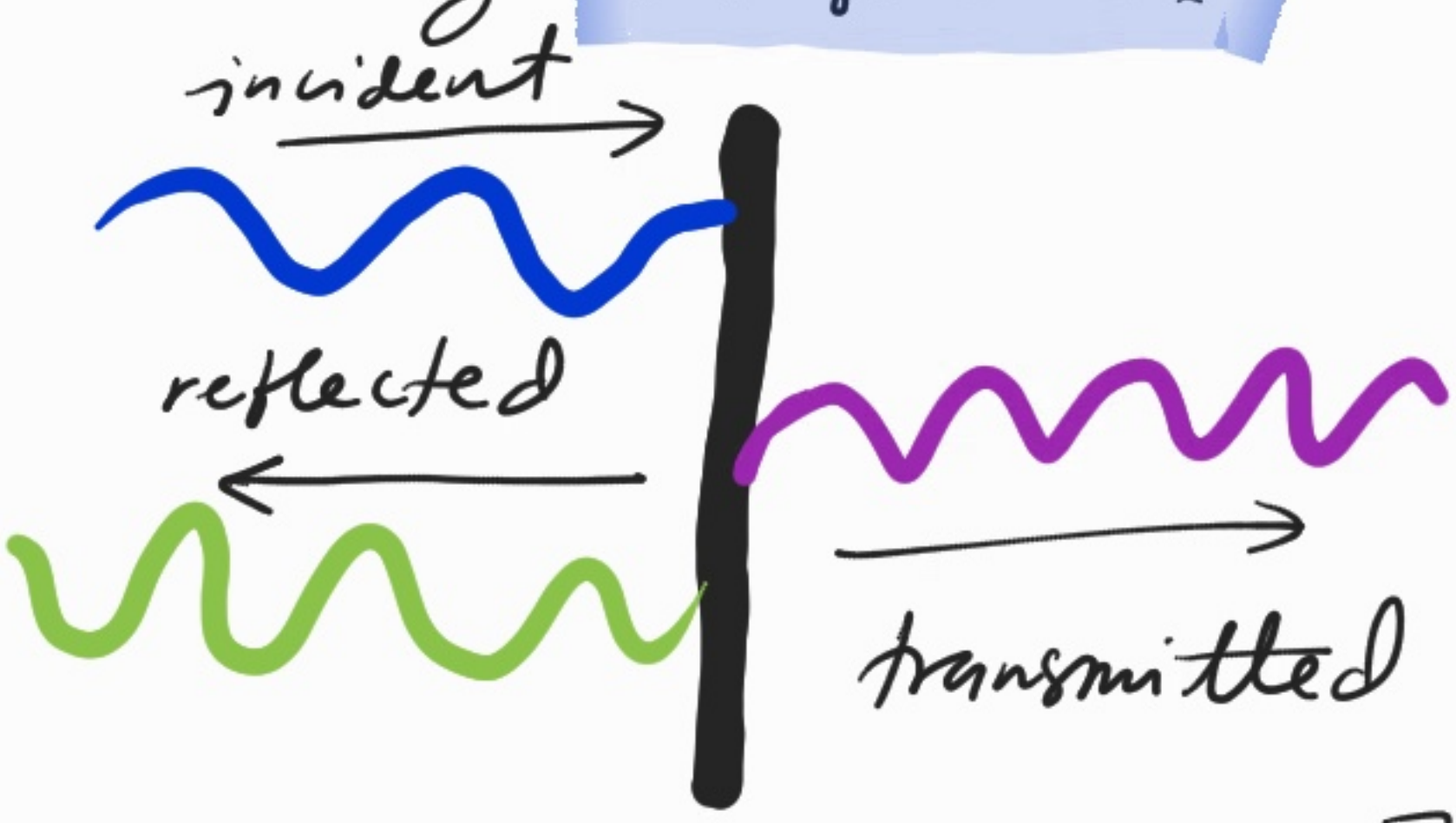
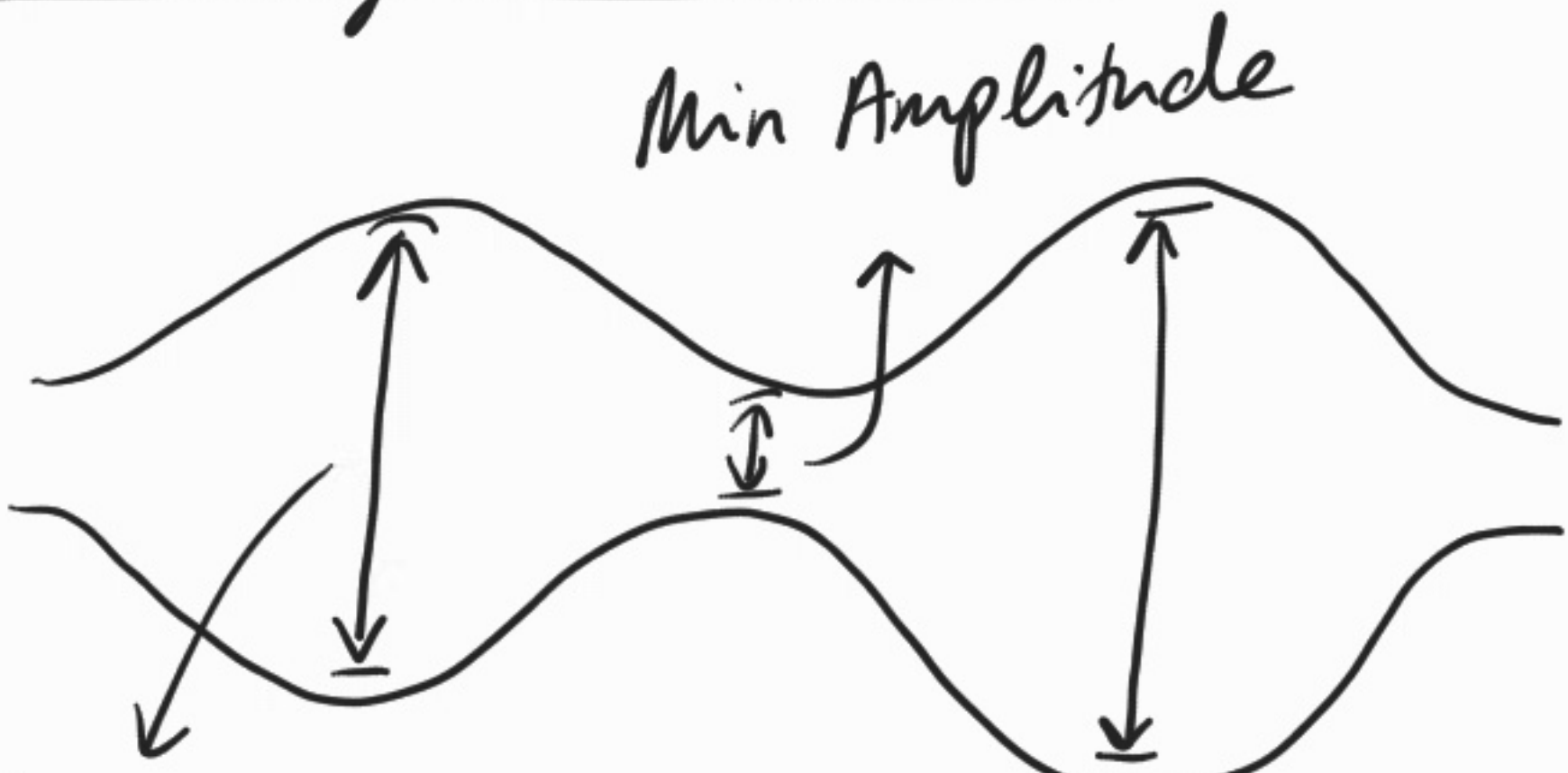
Standing Wave Ratio

- Continuous travelling wave is incident on a boundary and a continuous reflection travels back through the incoming wave - creating interference.



[single frame / instance]

Standing Wave Ratio



- Standing wave ratio [SWR]

$$SWR = \left[\frac{y_{m,i} + y_{m,r}}{y_{m,i} - y_{m,r}} \right]$$

$y_{m,i}$ – amplitude of incident wave

$y_{m,r}$ – amplitude of reflected wave.

Standing Wave Ratio

- SWR in terms of reflection coefficients:

$$SWR = \frac{1 + \left[\frac{y_{m,r}}{y_{m,i}}\right]}{1 - \left[\frac{y_{m,r}}{y_{m,i}}\right]} = \left[\frac{1+r}{1-r}\right]$$

- Total reflection:
  As $r \to 1$, $SWR \to \infty$

- No reflection:
  As $r \to 0$, $SWR \to 1$

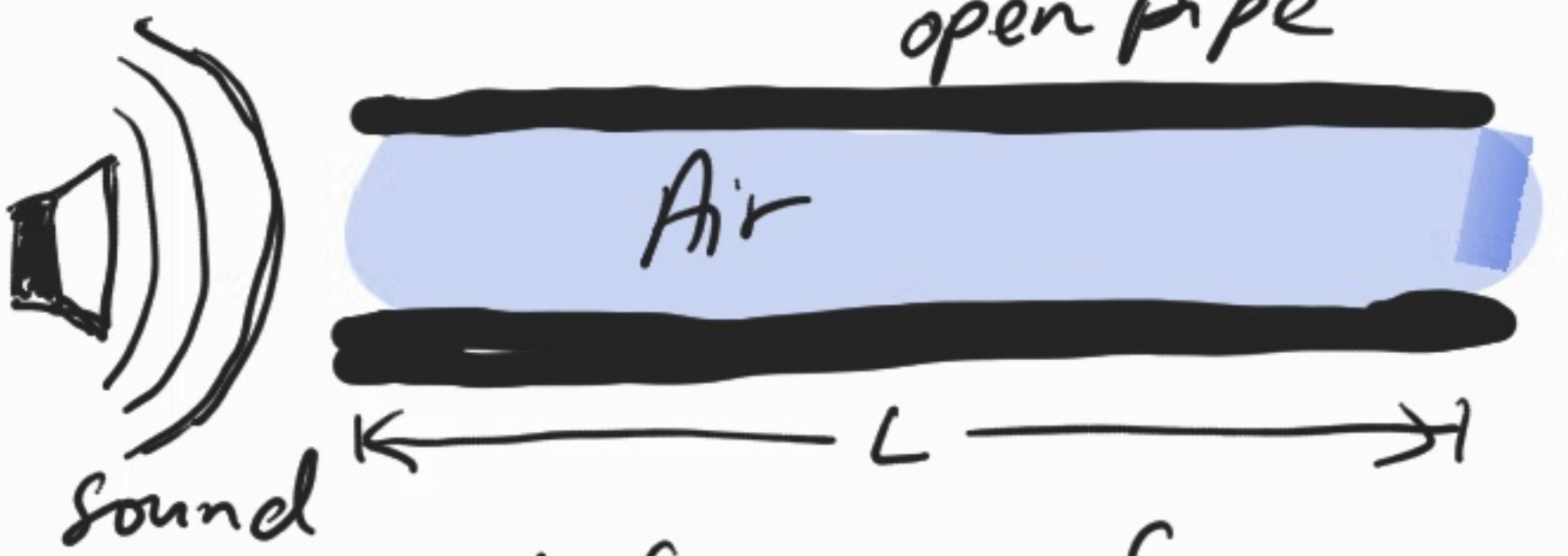
Standing Wave Ratio

- SWR in terms of power reflection coefficients:

$$R = r^2 \longrightarrow r = \sqrt{R}$$

$$SWR = \frac{1+\sqrt{R}}{1-\sqrt{R}}$$

Wave Resonance

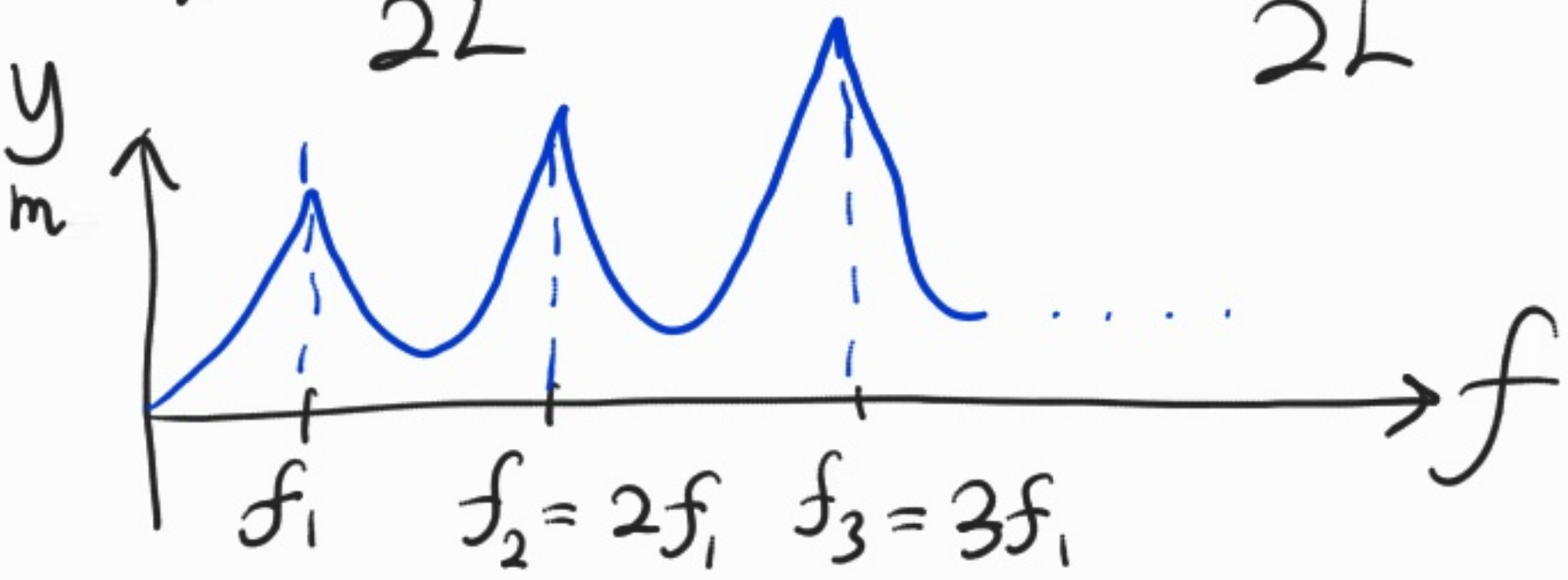


wave at frequency, f

⇒ Max amplitude occurs if the frequency of the speaker = normal mode frequencies of the pipe

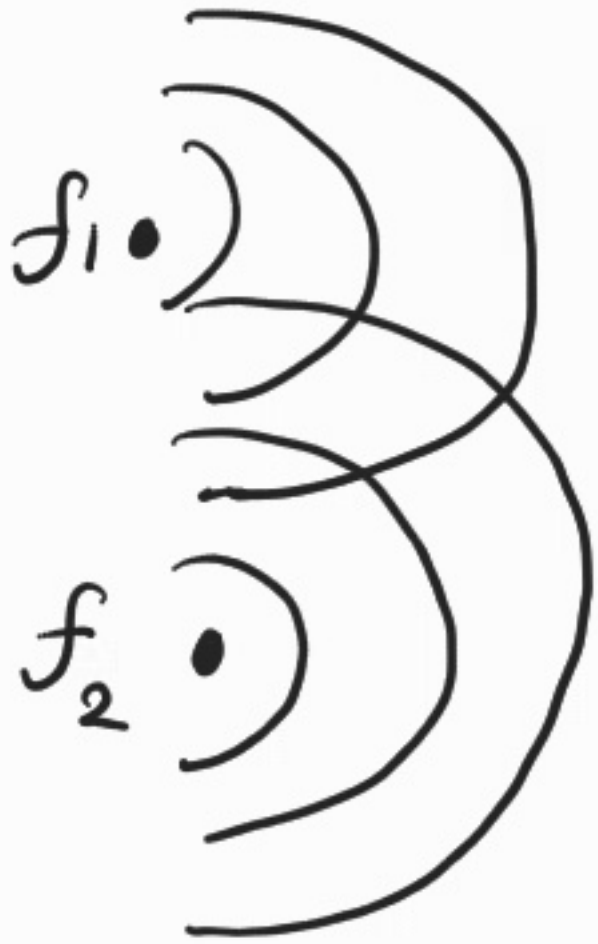
⇒ Open pipe wave harmonics :

$$f_n = \frac{nV}{2L} = nf \rightarrow f_1 = \frac{V}{2L}$$



Beats

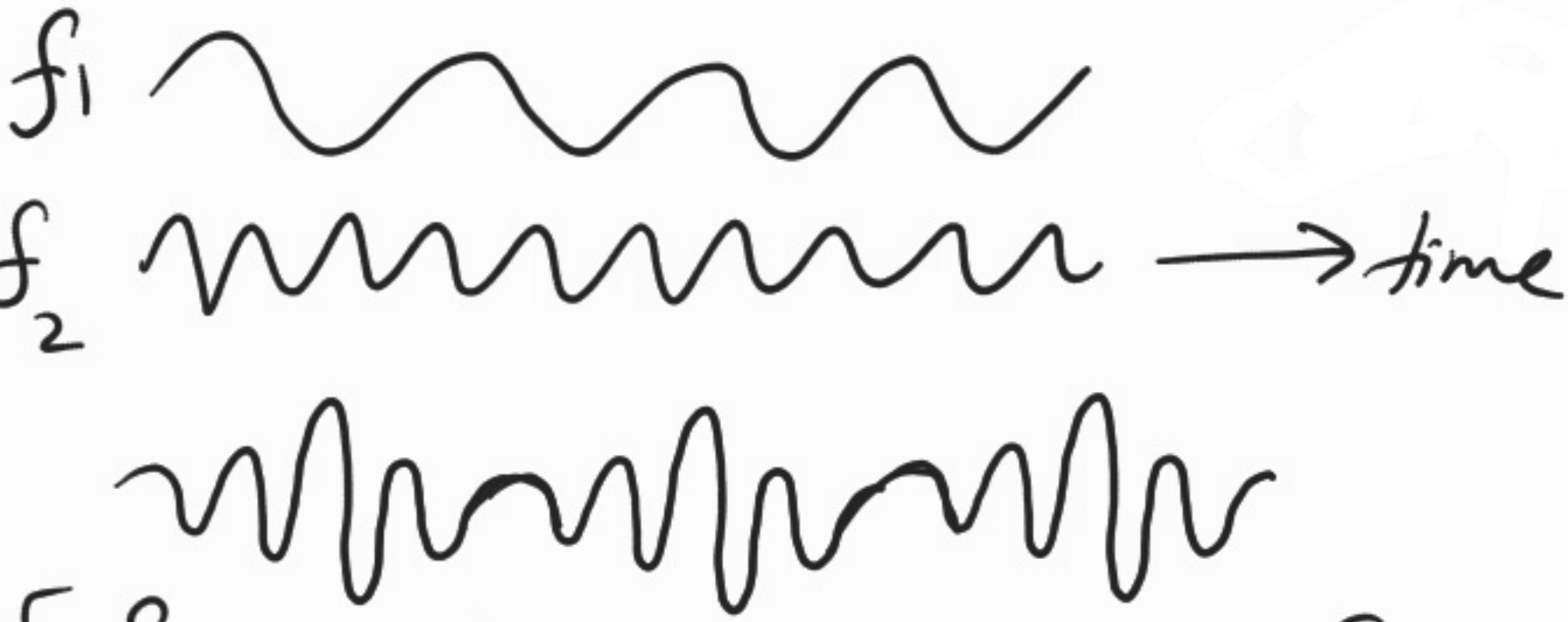
- Superposition of 2 harmonic waves with different frequencies:





} the resultant sound rises & falls in intensity forming beats.

- 2 waves with <u>same amplitude</u> but <u>different frequencies</u>.



[Superposition of waves] f_1 & f_2

Beats

- Consider 2 sound waves at a specific position – i.e. only time-dependent.

$$S_1 = S_m \cos(\omega_1 t)$$

$$S_2 = S_m \cos(\omega_2 t)$$

- Resultant Wave:

$$S_R = S_1 + S_2 = S_m \left[\cos(\omega_1 t) + \cos(\omega_2 t)\right]$$

$$\cos\alpha + \cos\beta = 2\cos\left[\frac{\alpha-\beta}{2}\right]\cos\left[\frac{\alpha+\beta}{2}\right]$$

$$S_R = 2 S_m \cos\left[\frac{1}{2}(\omega_1-\omega_2)t\right]\cos\left[\frac{1}{2}(\omega_1+\omega_2)t\right]$$

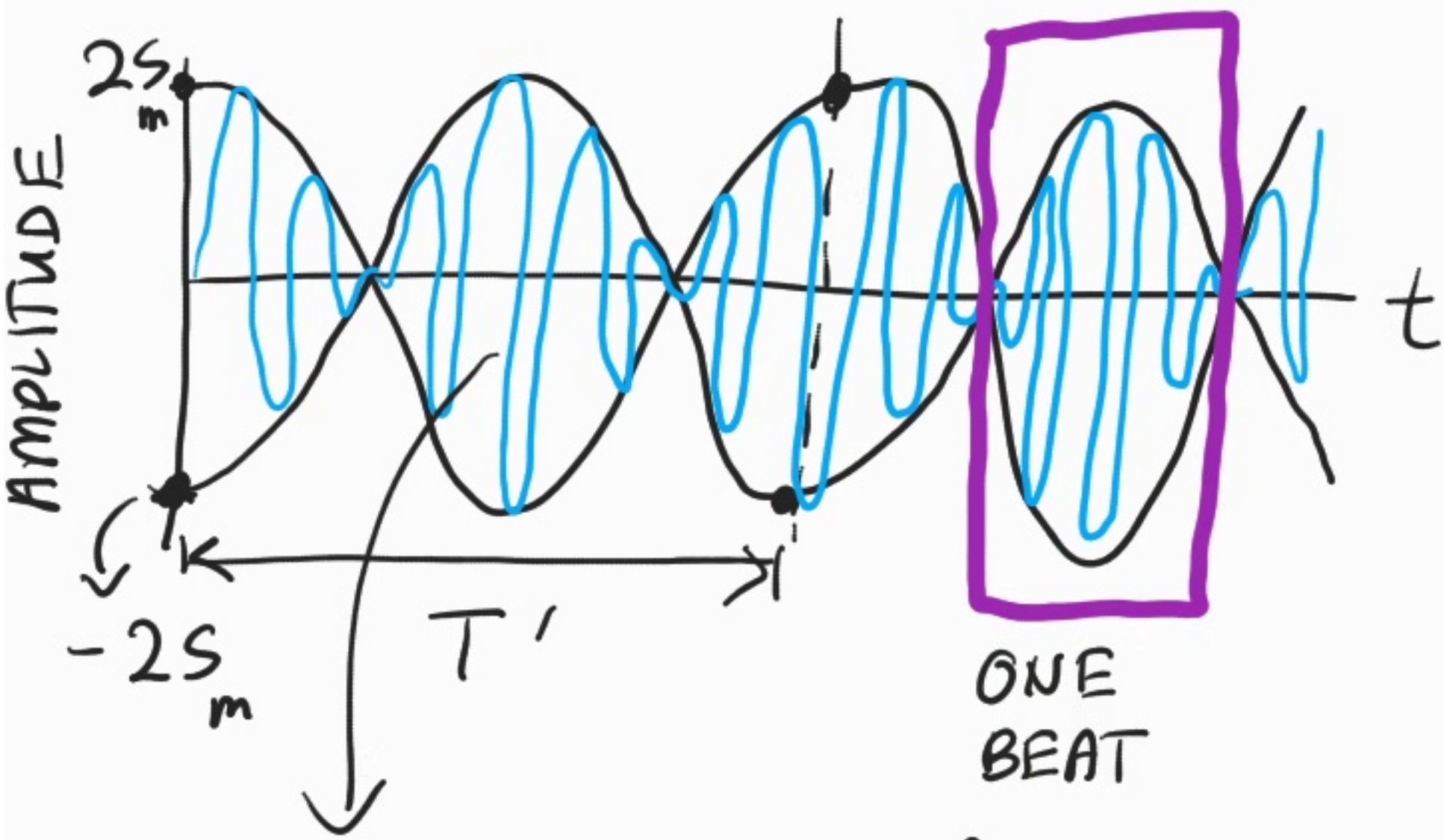
$$= \underbrace{\left[2 S_m \cos(\omega' t)\right]}\cos(\omega t)$$

$S'_{R,m}$ → time-dependent amplitude

$$\omega' = \frac{1}{2}(\omega_1-\omega_2)$$

$$\omega = \frac{1}{2}(\omega_1+\omega_2)$$

Beats



- wave oscillates with frequency :

$$f = \frac{f_1 + f_2}{2}$$

$$T' = \frac{T_1 - T_2}{2}$$

Beats

- The period of the time-dependent amplitude:

$$S_{R,M}(t) = 2S_M \cos(\omega' t)$$

$$\Rightarrow T' = \left[\frac{T_1 - T_2}{2}\right]$$

- Thus 1 beat occurs at:

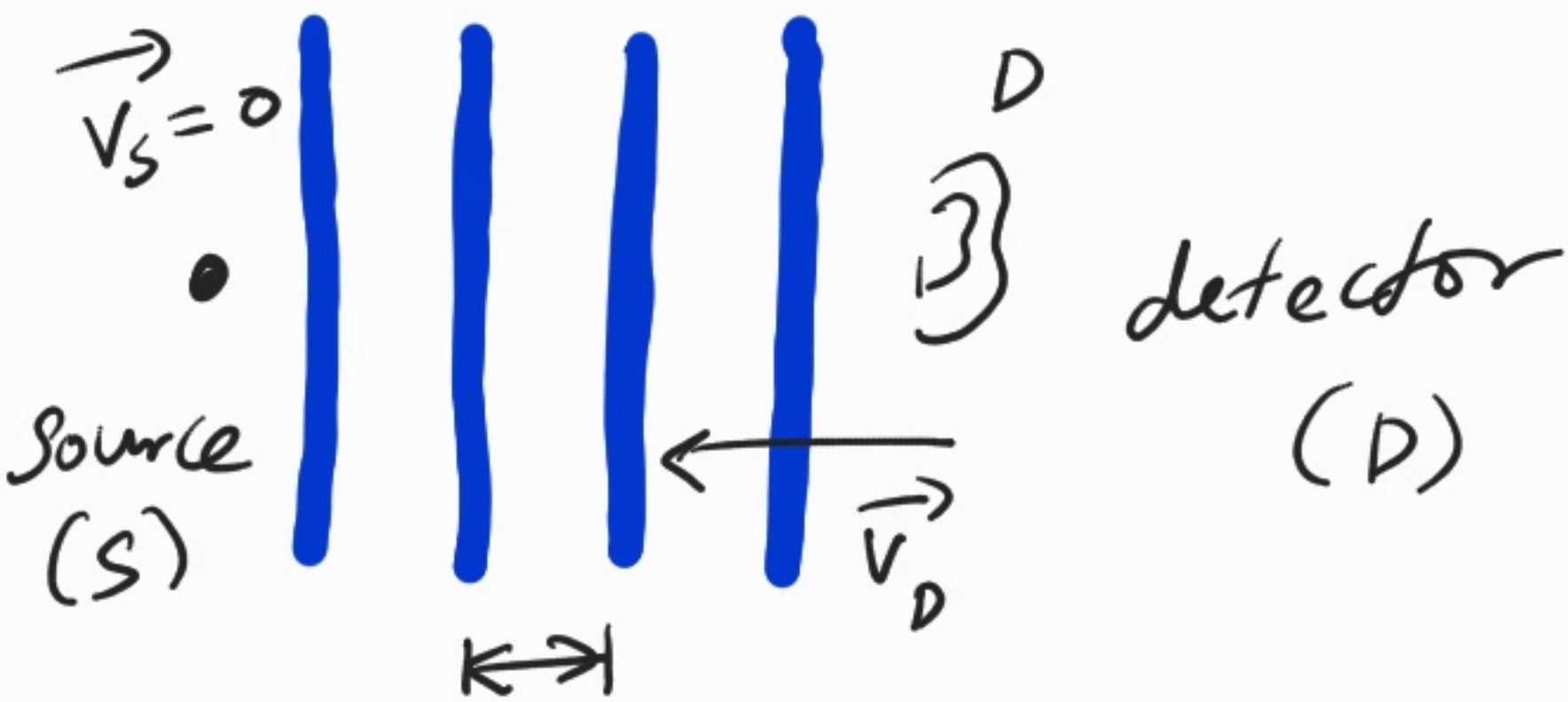
$$T_{beat} = \frac{T'}{2}$$

- Beat frequency $\Rightarrow$ $\left[\frac{\text{number of beats}}{\text{time taken}}\right]$

$$f_{beat} = 2f' = f_1 - f_2$$

Doppler Effect

- motion related to frequency change.



$\vec{V}_S$ → source velocity (m/s)

$\vec{V}_D$ → detector velocity (m/s)

v → wave speed.

Doppler Effect.

Definitions :

$vt \Rightarrow$ distance travelled by the wavefront in time, t.

$v_D t \Rightarrow$ distance the detector moves toward the source in time, t.

$(vt + v_D t) \Rightarrow$ distance travelled by the wavefront relative to the detector in time, t.

$\left[\frac{vt + v_D t}{\lambda}\right] \Rightarrow$ the number of wavefronts intercepted by the detector.

Doppler Effect.

The rate (or frequency) that the detector intercepts the wavelengths:

$$f' = \frac{\left[\frac{vt + v_D t}{\lambda}\right]}{t} = \left[\frac{v + v_D}{\lambda}\right]$$

Since $\lambda = \frac{v}{f}$

$$\therefore \quad f' = f\left[\frac{v + v_D}{v}\right] \#$$

Doppler Effect

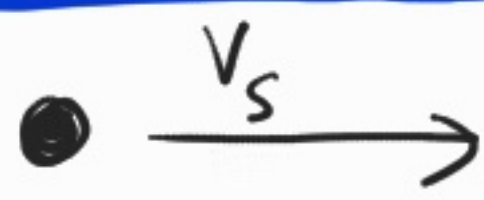
- General Doppler Effect Equation:
- f is the frequency intercepted by the detector if both detector and source are stationary.

$$f' = f\left[\frac{v \pm v_D}{v \pm v_s}\right]$$

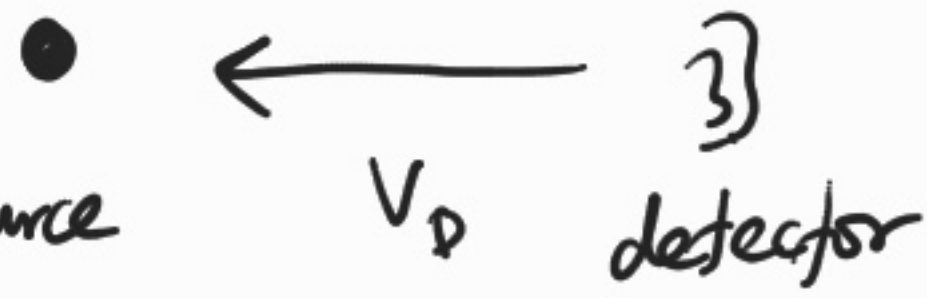
- Note: Sign Convention.

⇒ Signs are chosen such that the f' tends to be greater for motion towards and less for motion away (between source & detector).

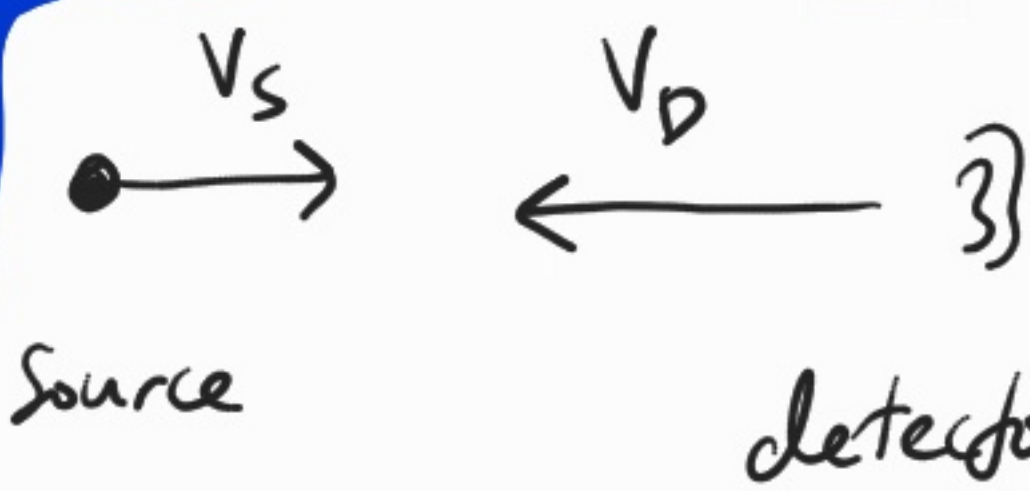
Doppler Effect



$$f' = f\left[\frac{v}{v - v_s}\right]$$



$$f' = f\left[\frac{v + v_D}{v}\right]$$



$$f' = f\left[\frac{v + v_D}{v - v_s}\right]$$

Wave Coherence

- 2 waves are coherent if:
  a) They are <u>monochromatic</u>
     [single frequency].
  b) There is a constant phase relation between them
     [in/out of phase or something in between]

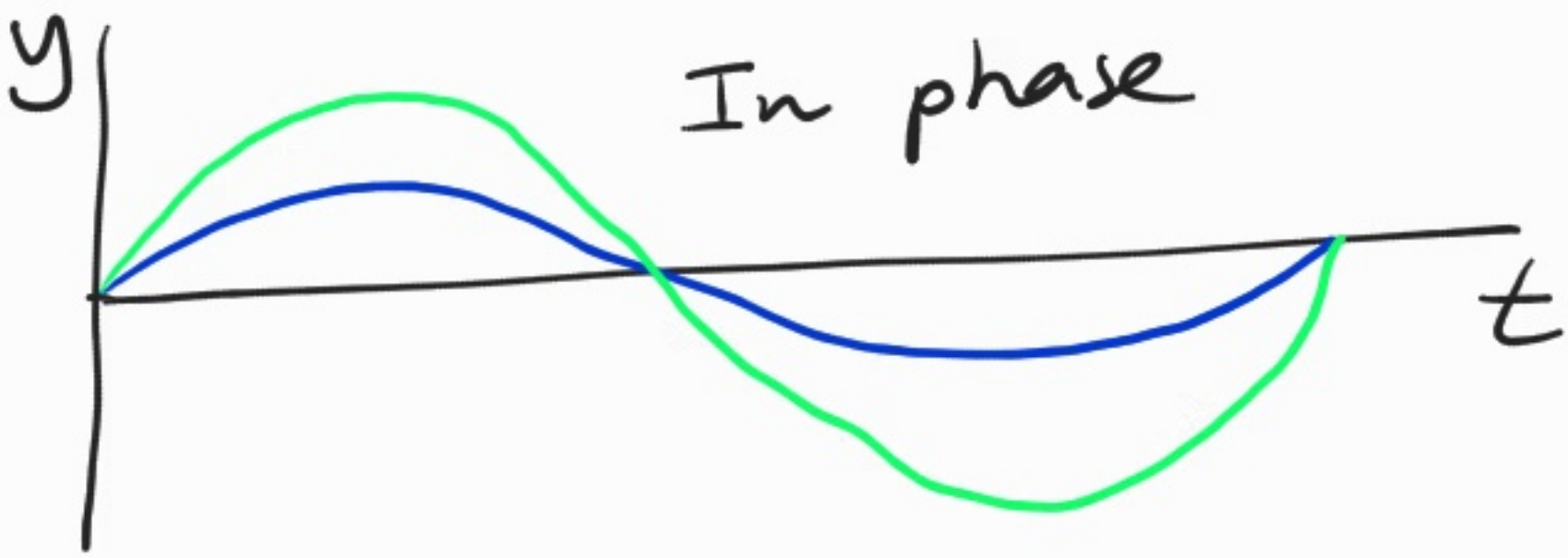
- Consider 2 coherent waves

$$y_1(x,t) = y_{m,1}\cos(kx - \omega t + \phi_1)$$

$$y_2(x,t) = y_{m,2}\cos(kx - \omega t + \phi_2)$$

Wave Coherence

Case I: In phase & Single Frequency



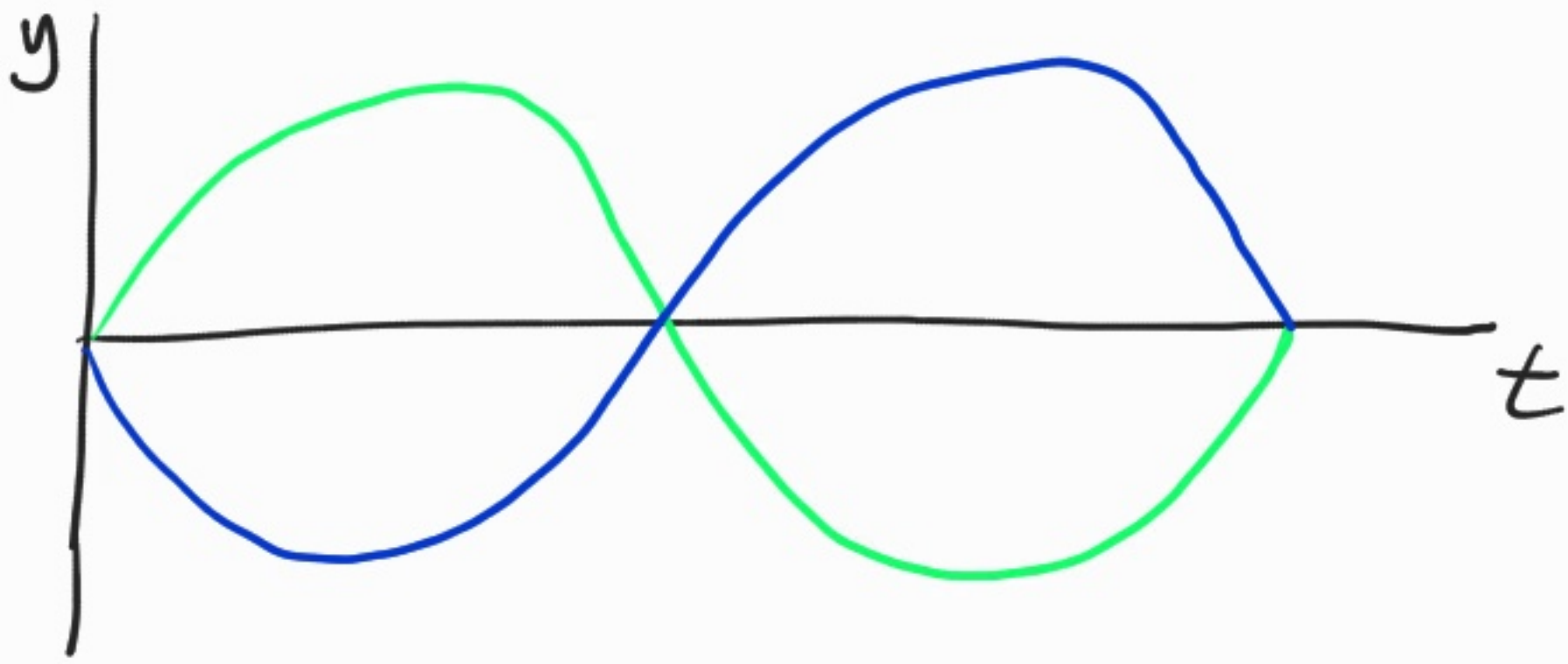
$$kx - \omega t + \phi_1 = kx - \omega t + \phi_2$$

$$\therefore \quad \phi_1 - \phi_2 = 0 + 2n\pi$$

$$\boxed{\phi_1 - \phi_2 = 2n\pi}$$

Wave Coherence

Case II : Out of Phase



$$\phi_1 - \phi_2 = (2n+1)\pi$$

Coherence

- To have interference pattern on a screen, light waves have to be coherent.
- The phase difference must be time-independent.

(1) Coherent Laser light:



(2) Incoherent light

